

1. BRAIN: OBJECTIVE PROPOSAL

Archive-conditioned prompt packet Π_g

Interface +
benchmark context

parent, champion
+ diverse elites

metric
deltas, trajectories +
failures

strict candidate schema

LLM proposer $q_\lambda(\cdot | \Pi_g)$
stateless provider call

Mutation:
rewrite | recombine | simplify | repair

Sample
response $y_{g,i}$

Raw candidate objective $y_{g,i}$

$$f_\theta(x; d) = f_{\text{base}}(x; d) + \sum_j \alpha_j h_j(T(x; d))$$

Search artifact:
differentiable objective

2. TOOLS: SAFE EXECUTION AND EVALUATION

Safe lowering L_G

AST +
grammar

ObjectiveSpec
lowering

finite-value +
autograd gate

parse only,
never execute directly

Validated
Objective-Spec
 $s_{g,i}$

COST-SCALED EVALUATION

Tier A – DREAMPlace

all validated candidates
HPWL, density, gradients

valid collateral
+admitted proxy

Tier B - Timing proxy

eligible candidates
WNS/TNS signal

selected finalist

Tier C - OpenROAD

selected finalists
routing-stage evidence

$L_G(y_{g,i}) = \perp$ or unsafe

available execution evidence $m(\cdot)$

**Score +
Archive
update**

$S(f, m)$

cell key $\phi(f)$

retain best
objective in
each cell

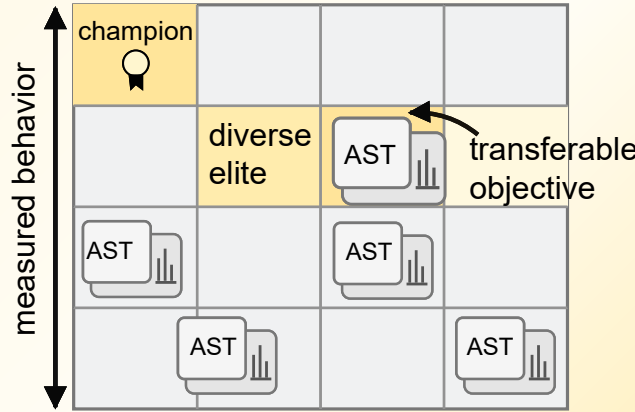
insert or
replace
elite

**Negative
memory**

parse failures
gradient failures
HPWL/density
regressions
...

MAP-Elites archive A_{g+1}

objective mechanism



champion
+diverse elites

+ lineage
neighbors

Feedback reconstruction

Retrieve H_{g+1}

failure
reasons+
repair
notes

rank and metric deltas
champion and diverse
elites

component trajectories
relevant failure lessons

ranks + metric deltas + component trajectories + failure lessons